

Supplementary Information

Approach-Region Dynamics and Rate Turnover in Thermally Activated Barrier Crossing

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S1 Mathematical Foundation

The results in this section establish the dynamical structure of barrier crossing used throughout the manuscript. The treatment follows the standard reaction-dynamics formulation of dissipative barrier crossing developed in the Kramers, Pollak–Grabert–Hänggi, and Grote–Hynes frameworks, with the damping ratio $\chi = \Gamma/(2\Omega)$ used here as the organizing dimensionless parameter.

S1.1 The exceptional point at $\chi = 1$

Note: This section refers exclusively to the stable precursor-well generator ($\Omega_0^2 > 0$), not the inverted saddle-point coordinate. The EP2 structure derived below belongs to the approach-dynamics $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$; no EP exists at the inverted barrier (see Section ?? of the main text).

The second-order generator $\ddot{q} + \Gamma\dot{q} + \Omega^2 q = 0$ is equivalent to the first-order system $d\mathbf{x}/dt = A\mathbf{x}$ with state vector $\mathbf{x} = (q, \dot{q})^\top$ and matrix

$$A = \begin{pmatrix} 0 & 1 \\ -\Omega^2 & -\Gamma \end{pmatrix}. \quad (\text{S1})$$

The eigenvalues are $\lambda_{\pm} = -\Gamma/2 \pm \sqrt{\Gamma^2/4 - \Omega^2}$. For $\chi \neq 1$ these are distinct and A is diagonalizable over $\mathbb{C}$. At $\chi = 1$, eigenvalues coalesce at $\lambda_+ = \lambda_- = -|\Omega|$, the geometric multiplicity drops to 1, and A is defective with Jordan normal form

$$J = \begin{pmatrix} -|\Omega| & 1 \\ 0 & -|\Omega| \end{pmatrix}, \quad (\text{S2})$$

which is the defining signature of a second-order exceptional point in the damped generator governing the barrier-crossing dynamics [17]. The time-domain Green's function changes class at this point:

$$\chi < 1: \quad h(t) = e^{-\Gamma t/2} \sin(\Omega_d t) / \Omega_d \quad (\text{oscillatory; recrossing}) \quad (\text{S3})$$

$$\chi = 1: \quad h(t) = t e^{-|\Omega|t} \quad (\text{polynomial-exponential; critical}) \quad (\text{S4})$$

$$\chi > 1: \quad h(t) = C_+ e^{\lambda_+ t} + C_- e^{\lambda_- t} \quad (\text{biexponential; overdamped}) \quad (\text{S5})$$

The long-time decay rate for $\chi > 1$ is controlled by the less-negative eigenvalue:

$$|\lambda_-| = |\Omega| \left(\chi - \sqrt{\chi^2 - 1} \right). \quad (\text{S6})$$

The function $f(\chi) = \chi - \sqrt{\chi^2 - 1}$ is strictly decreasing for $\chi > 1$ with $f(1) = 1$ and $f(\chi) \rightarrow 0$ as $\chi \rightarrow \infty$. Among all parameter choices producing purely monotone decay, $\chi = 1$ therefore maximizes the decay rate: fastest commitment without oscillation.

S1.2 The dynamical efficiency curve and the 5% window

The change in solution character at $\chi = 1$ has a direct kinetic consequence: in barrier-crossing theory the competition between energy diffusion and spatial diffusion produces a turnover in the reaction rate as friction varies.

The rate turnover is described by a PGH-like harmonic-mean interpolation between the two limiting regimes:

$$k(\chi) = [k_{\text{ED}}^{-1}(\chi) + k_{\text{SD}}^{-1}(\chi)]^{-1} \times k_{\text{TST}} \quad (\text{S7})$$

where the energy-diffusion (low-friction) branch satisfies $k_{\text{ED}} \propto \chi$ (rate rises with friction as the bath supplies activation energy) and the spatial-diffusion (high-friction) branch satisfies $k_{\text{SD}} \propto 1/\chi$ (rate falls with friction due to viscous drag). For this symmetric schematic, the harmonic-mean interpolation gives exactly:

$$k(\chi) \propto \frac{\chi}{1 + \chi^2}, \quad \eta(\chi) = \frac{2\chi}{1 + \chi^2}, \quad (\text{S8})$$

which has an exact maximum at $\chi = 1$ and a 95% window of $\chi \in [0.72, 1.38]$ (roots of $\eta(\chi) = 0.95$, computed independently of any dataset).

Derivation of the operational window. The $[0.8, 1.3]$ near-critical window used throughout this work is derived from two criteria applied to the symmetric schematic Eq. (S8):

$$\eta(0.8) = \frac{2(0.8)}{1 + (0.8)^2} = \frac{1.6}{1.64} = 0.976 \geq 0.95 \quad (\text{S9})$$

$$\eta(1.3) = \frac{2(1.3)}{1 + (1.3)^2} = \frac{2.6}{2.69} = 0.967 \geq 0.95 \quad (\text{S10})$$

Both boundaries satisfy the stated $\eta \geq 0.95$ criterion. The window $[0.8, 1.3]$ is therefore a mathematically grounded subset of the schematic 95% band $[0.72, 1.38]$, with the 0.80 lower bound and 1.30 upper bound chosen as conservative round-number boundaries relative to the schematic criterion. The K-promoted Fe(111) ground-state estimate $\chi_0 \approx 1.28$ independently satisfies $\eta(1.28) = 0.970 \geq 0.95$; it is within the window by the derived criterion and was not used to set the boundary.

Note on the spatial-diffusion branch. A more physical choice is $k_{\text{SD}} \propto \kappa_{\text{Kr}}(\chi) = \sqrt{1 + \chi^2} - \chi$. Substituting into the harmonic mean gives $k \propto \chi - \chi^2/\sqrt{1 + \chi^2}$, which peaks near $\chi \approx 0.79$, not exactly at 1. Figure 1 uses the symmetric schematic as a conceptual illustration; the $[0.8, 1.2]$ window from the full Mel'nikov–Meshkov/PGH formula at $V^\ddagger/k_B T \geq 8$ (Pollak, Grabert & Hänggi 1989 [6]; Pollak & Talkner 2005 [3]) provides a complementary conservative estimate. The operational $[0.8, 1.3]$ window lies between the conservative PGH estimate and the schematic 95% band.

Figure 1 uses the normalized form $\eta(\chi) = k(\chi)/k_{\text{max}}$ from the symmetric schematic Eq. (S8). This is a conceptual illustration; exact window boundaries for specific systems require the full PGH numerical evaluation referenced above.

A note on the Kramers transmission coefficient: the Kramers–Smoluchowski coefficient in the high-friction limit is $\kappa_{\text{Kr}} = \sqrt{1 + \chi^2} - \chi$, which is strictly monotonically decreasing from 1 to 0 and has no maximum. The Kramers turnover arises from the combined PGH rate (energy-diffusion plus spatial-diffusion regimes), not from κ_{Kr} alone. κ_{Kr} should not be presented as having a maximum at $\chi = 1$.

S1.3 Connection to Kramers, Grote–Hynes, and Zusman

Grote–Hynes extension. In non-Markovian baths, the effective χ becomes frequency-dependent: $\chi_{\text{eff}}(\Omega) = \text{Re}[\hat{\gamma}(\Omega)]/(2\Omega_0)$, where $\hat{\gamma}(\Omega)$ is the Laplace-transformed memory kernel. For frictional dissipation, the physically appropriate quantity is the *dissipative* (real) component of the Debye-type response:

$$\text{Re}[\hat{\gamma}(\Omega)] = \frac{\gamma_0}{1 + (\Omega\tau_{\text{mem}})^2}. \quad (\text{S11})$$

The response magnitude $|\hat{\gamma}| = \gamma_0/\sqrt{1 + (\Omega\tau_{\text{mem}})^2}$ is larger but includes the reactive (non-dissipative) component. Both reduce to γ_0 in the Markovian limit $\Omega\tau_{\text{mem}} \rightarrow 0$. The $\chi_0 = 1$ separatrix broadens into a band of width $\sim \Omega\tau_{\text{mem}}$ but remains well-defined for $\Omega\tau_{\text{mem}} < 0.5$.

Zusman theory. The adiabaticity parameter $g = 2\pi V^2\tau_L/(\hbar\sqrt{4\pi\lambda k_B T})$ scales as $g \sim 2\chi_{\text{ET}}$ to within a prefactor that evaluates to 0.67–1.0 at ambient temperature for $\lambda \in [0.5, 1.0]$ eV. These two quantities are distinct and must not be conflated:

Quantity	Physical meaning	Tunes
g (Zusman)	Electronic coupling adiabaticity	Role of H_{ab} (electronic coupling matrix element)
χ	Solvent friction damping	Role of τ_L (nuclear dynamics)

Table S1: Distinction between the Zusman adiabaticity parameter g and the damping ratio χ . Both conditions must be satisfied simultaneously for efficient concerted electron transfer.

The mapping $g \sim 2\chi$ is specific to ambient-temperature ET with standard reorganization energies. At cryogenic temperatures or with very small or large λ , the full temperature-dependent expression is required.

S2 Detailed Bias Analysis by Estimator Route

S2.1 Route A: spectroscopic proxy

The connection between observed linewidth and the true dissipation rate is:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_2^*} \implies \Delta\tilde{\nu}_{\text{obs}} = \underbrace{\Delta\tilde{\nu}_L}_{\text{homogeneous}} + \underbrace{\Delta\tilde{\nu}_G}_{\text{inhomogeneous}} \quad (\text{S12})$$

Route A yields $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$, which equals the true damping ratio of the measured vibrational mode when $T_2 \approx 2T_1$ (homogeneous lifetime broadening dominates). In precursor-state measurements this is an approach-region estimate; it equals χ_{TS} only if the measured mode independently corresponds to the saddle-region normal mode.

The χ_{spec} values cited for Fe(111) and Fe(110) in N_2 activation derive from DFT-computed electronic friction linewidths Γ_{cat} (Maurer et al. 2017), not from raw operando IR linewidths; these are Route C inputs interpreted as Route A estimates. $\chi_{\text{spec}} \approx \chi_{\text{DFT}}$ only when homogeneous broadening dominates the measured spectrum, a condition that must be verified experimentally. Until operando T_1 measurements are available for these systems, the reported χ_{spec} values carry the Route C bias described in Section S2.3.

Table S2 summarizes sources of linewidth inflation and their mitigations.

Source	Magnitude	Diagnostic	Mitigation
Thermal Doppler broadening (surface)	20–50 cm ⁻¹ at 400°C	Gaussian profile component	Measure at low T ; extrapolate Lorentzian to $T \rightarrow 0$
Inhomogeneous site distribution	Varies with surface prep	Coverage-dependent linewidth	Single-crystal at low coverage; verify $\Delta\tilde{\nu}$ vs. θ
Pure dephasing (T_2^*)	Coupling-dependent	Voigt Gaussian fraction > 50%	Extract Lorentzian component only; cross-check with T_1
Instrument lineshape	Apodization artifact	Deviation from Lorentzian	Deconvolve instrument function; high-resolution FTIR

Table S2: Sources of linewidth inflation in Route A and their mitigations.

S2.2 Route B: uncertainty propagation

Under the corrected Debye-Drude model, the absolute chi estimate is $\chi_{\text{ET}} = \Omega_s^2 \tau_L / (2\Omega_0)$, where $\Omega_0 = \sqrt{2\lambda / (m_{\text{eff}} \Delta Q^2)}$. The propagation of m_{eff} uncertainty follows from:

$$\Omega_0 \propto m_{\text{eff}}^{-1/2} \implies \chi_{\text{ET}} \propto m_{\text{eff}}^{+1/2} \quad (\text{S13})$$

A decade uncertainty in m_{eff} (50 to 500 amu) shifts χ_{ET} by a factor of $\sqrt{10} \approx 3.2$. Additional uncertainty in Ω_s (50–150 cm⁻¹) contributes a further factor of ~ 3 , and the two uncertainties compound. Table S3 illustrates this with exact Python-computed values for $\tau_L = 1.2$ ps, $\lambda = 0.7$ eV, $\Delta Q = 0.5$ Å:

m_{eff} (amu)	Ω_s (cm ⁻¹)	χ_{ET} (absolute)
50	100	6.5
100	100	9.2
200	100	13.0
500	100	20.5
100	50	2.3
100	150	20.6

Table S3: Route B absolute χ_{ET} (Python-computed) for propylene carbonate ($\tau_L = 1.2$ ps) across a decade of m_{eff} and Ω_s . All values are deeply overdamped and span a 9-fold range, demonstrating that absolute Route B estimates are not reliable for regime classification. Absolute values are not used for inference in this work; all conclusions rest on relative χ only.

For relative χ within a homologous solvent series, m_{eff} , λ , and ΔQ cancel exactly for a fixed donor–acceptor pair. $C_{\text{solv}} = \Omega_s^2$ cancels approximately when the compared solvents have similar librational frequencies; for similar-class solvents (e.g. nitriles or carbonates), the approximation $\chi_{\text{ET}}^{(A)} / \chi_{\text{ET}}^{(B)} \approx \tau_L^{(A)} / \tau_L^{(B)}$ introduces $\lesssim 40\%$ error. For cross-class comparisons, the full corrected ratio $(\Omega_{s,A}^2 / \Omega_{s,B}^2) \times (\tau_L^{(A)} / \tau_L^{(B)})$ should be used. Table S4 gives the relative series used throughout the main text.

Solvent	τ_L (ps)	χ_{rel} (χ/χ_{MeCN})
Acetonitrile	0.20	1.0 (reference)
Water	0.20	1.0
D ₂ O	0.30	1.5
THF	0.80	4.0
Propylene carbonate	1.20	6.0

Table S4: Relative damping ratios for the TEMPO solvent series. The monotonic rate decrease with increasing τ_L is consistent with sampling the high-friction (spatial-diffusion) side of the turnover; the near-critical optimum is predicted at $\tau_L < 0.20$ ps, outside the sampled range. Absolute flank assignment requires an independently calibrated χ anchor.

S2.3 Route C: DFT friction bias

The Newns–Anderson expression $\Gamma_{\text{cat}} = 2\pi|V_k|^2\rho(\varepsilon_F)$ is a mean-field, single-particle result [18,19]. Three sources of systematic error require attention.

For 3d ferromagnets, GGA-DFT underestimates d-band exchange splitting in Fe, Co, and Ni. The predicted Γ_{cat} can be off by 20–50%. For quantitative predictions on these metals, DFT+ U with $U \approx 2\text{--}4$ eV for Fe is recommended; Route A provides a model-independent cross-check.

Route C computes χ at the adsorbate ground-state geometry and is therefore a lower bound on χ_{TS} : as π^* -backdonation increases and Ω_{bond} decreases as the bond stretches toward the transition state, χ_{DFT} understates χ_{TS} in most catalytic systems. For quantitative transition-state predictions, Γ_{NN} and the harmonic frequency should be recomputed at the saddle-point geometry.

Spin-orbit coupling is negligible for light adsorbates (N, O, CO) on 3d metals but becomes relevant for heavy adsorbates on Pd, Pt, and Au; non-collinear DFT should be included if the adsorbate is heavy.

Cross-check: Γ_{cat} is an energy in eV. The corresponding electronic hybridization linewidth is $\Delta\tilde{\nu} = \Gamma_{\text{cat}}/(hc)$ in cm^{-1} . This electronic width probes a different dissipation channel than the vibrational T_1/T_2 lifetime measured by Route A; exact agreement with $\Delta\tilde{\nu}_L$ is not expected. Order-of-magnitude consistency (within a factor of ~ 10) confirms no gross discrepancy between electronic and vibrational friction scales.

S3 Domain-Specific Derivations

This section shows how the same damped barrier-crossing generator appears in three chemically distinct reaction classes—electron transfer, proton-coupled electron transfer, and heterogeneous surface reactions—demonstrating that the dimensionless damping ratio χ can be estimated from experimentally or computationally accessible observables in each case.

S3.1 Electron transfer: mapping to Marcus–Zusman

The Marcus solvent reorganization coordinate Q obeys:

$$m\ddot{Q} + \gamma_{\text{solv}}\dot{Q} + k(Q - Q_{\text{min}}) = \xi(t), \quad k = \frac{2\lambda}{(\Delta Q)^2} \quad (\text{S14})$$

In a Debye-Drude polar solvent, the friction coefficient scales as $\gamma_{\text{solv}} = m\Omega_s^2\tau_L$ (van der Zwan & Hynes

1982), where Ω_s is the characteristic solvent librational frequency. Identifying terms with the stable precursor-well generator: $\Omega_0^2 = k/m = 2\lambda/(m_{\text{eff}}\Delta Q^2)$; $\Gamma_{\text{eff}} = \gamma_{\text{solv}}/m = \Omega_s^2 \tau_L \equiv C_{\text{solv}} \tau_L$. Therefore:

$$\chi_{\text{ET}} = \frac{C_{\text{solv}} \tau_L}{2\Omega_0} = \frac{\Omega_s^2 \tau_L}{2\sqrt{2\lambda/(m_{\text{eff}}\Delta Q^2)}} \quad (\text{S15})$$

Because $\chi \propto \tau_L$, slower solvents (larger τ_L) produce larger χ (more overdamped). The relative chi cancels C_{solv} , m_{eff} , λ , and ΔQ exactly for a fixed donor-acceptor pair: $\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$.

The Zusman adiabaticity parameter $g = 2\pi V^2 \tau_L / (\hbar \sqrt{4\pi \lambda k_B T})$. Using the corrected Route B definition $\chi_{\text{ET}} = \Omega_s^2 \tau_L / (2\Omega_0)$:

$$\frac{g}{2\chi_{\text{ET}}} = \frac{\pi V^2 \Omega_0}{\hbar \Omega_s^2 \sqrt{\pi \lambda k_B T}} \quad (\text{S16})$$

In the limiting case $\Omega_0 \approx \Omega_s$ (solvation-coordinate frequency identified with bath librational frequency), this simplifies to:

$$\frac{g}{2\chi_{\text{ET}}} \approx \frac{\pi V^2}{\hbar \Omega_s \sqrt{\pi \lambda k_B T}} \quad (\text{S17})$$

At $T = 300$ K, $\lambda = 0.7$ eV, $V = 200$ cm⁻¹, $\Omega_s = 100$ cm⁻¹, this prefactor evaluates to approximately 0.8, confirming $g \sim 2\chi$ to within order unity at standard conditions under the stated approximation. The mapping $g \sim 2\chi$ does not mean g and χ are the same quantity; see Section S1.3 for the distinction.

Why the Kramers turnover maximum structurally implies $\chi \approx 1$

The PGH combined rate operates in two opposing limits: the energy-diffusion branch gives $k_{\text{ED}} \propto \Gamma$ (rate rises with friction because the bath supplies activation energy) and the spatial-diffusion branch gives $k_{\text{SD}} \propto 1/\Gamma$ (rate falls with friction because viscous drag slows barrier transit). Their crossover defines the rate maximum. In the spatial-diffusion limit, the Kramers transmission coefficient for escape over a parabolic barrier is

$$\kappa_{\text{Kr}} = \sqrt{\left(\frac{\Gamma}{2\omega_b}\right)^2 + 1} - \frac{\Gamma}{2\omega_b}, \quad (\text{S18})$$

which is strictly monotone decreasing and convex for all $\chi > 0$ ($d\kappa_{\text{Kr}}/d\chi = \chi/\sqrt{\chi^2 + 1} - 1 < 0$ always; $d^2\kappa_{\text{Kr}}/d\chi^2 = (1 + \chi^2)^{-3/2} > 0$ always). κ_{Kr} has no maximum and no inflection point near $\chi = 1$; it is steepest at $\chi \rightarrow 0$.

The PGH rate maximum near $\chi \approx 1$ does not arise from κ_{Kr} alone. It arises from the competition between the energy-diffusion branch ($k_{\text{ED}} \propto \Gamma$, rising with friction) and the spatial-diffusion branch ($k_{\text{SD}} \propto 1/\Gamma$, falling with friction). Their crossover defines a rate maximum that the full PGH interpolation places near $\chi \approx 1$ for physically relevant barrier heights $V^\ddagger/k_B T \in [5, 30]$. Consequently $\chi \approx 1$ at the turnover is a structural consequence of the two-regime form of barrier-crossing theory, not of κ_{Kr} alone. Within the thermally activated, near-Markovian, single-dominant-coordinate regime considered here, systems exhibiting a

Kramers/PGH turnover are expected to place the rate maximum near order-unity χ , after the appropriate domain-specific mapping of Γ_{eff} and Ω_0 . This structural expectation does not require any particular chemistry; it follows from the two-regime form of barrier-crossing theory.

S3.2 Catalysis: Newns–Anderson to χ

An adsorbate on a metal surface has its vibrational level ε_a broadened by hybridization with metal continuum states:

$$\Gamma_{\text{cat}} = 2\pi|V_k|^2\rho(\varepsilon_F) \quad [\text{eV}] \quad (\text{S19})$$

Converting to a damping ratio comparable to the bond frequency:

$$\chi_{\text{DFT}} = \frac{\Gamma_{\text{int}} + \Gamma_{\text{cat}}}{2hc\tilde{\nu}_{\text{bond}}} \quad hc = 1.23984 \times 10^{-4} \text{ eV} \cdot \text{cm} \quad (\text{S20})$$

The K-promoter effect on Fe(111) operates simultaneously on both levers. K donates electron density, increasing $\rho(\varepsilon_F)$ and hence Γ_{cat} (raising Γ). Additional d-band backdonation into N_2 π^* reduces $\tilde{\nu}$ from 2359 cm^{-1} (gas phase) to 1415 cm^{-1} , decreasing Ω_{TS} (lowering Ω). Both effects increase χ toward unity, structurally accounting for the empirically observed activity enhancement.

S3.3 Fe(111) transition-state χ : analytical bracket

The Route C estimator applied to the ground-state geometry of K-promoted Fe(111) gives $\chi_{\text{DFT}} \approx 1.28$ using $\Gamma_{\text{NN}} = 0.45 \text{ eV}$ and $\tilde{\nu}_{\text{NN}} = 1415 \text{ cm}^{-1}$ (adsorbed-state frequency confirmed by HREELS). This section provides the analytical bracket showing that χ_{TS} is well into the overdamped regime at the saddle point, consistent with the approach-dynamics interpretation in the main text.

Two levers that shift χ between ground state and transition state

Two changes occur simultaneously on the path from ground state to transition state, and both push χ upward.

Bond softening (Ω_{TS} decreases). The N–N bond stretches from approximately $1.18 \text{ \AA}$ (adsorbed precursor, γ -state on Fe(111)) toward the transition-state geometry. The 6D-PES calculations of Nosir et al. place the most reactive dissociation channels at TS bond lengths of $1.5\text{--}1.8 \text{ \AA}$. Using Morse-potential scaling $\nu \propto \exp[-a(r - r_e)]$ calibrated to the known frequencies at r_e (gas phase: 2359 cm^{-1} at $1.10 \text{ \AA}$; γ -state: 1415 cm^{-1} at $1.18 \text{ \AA}$), the N–N stretch frequency at the TS geometry extrapolates to:

$$\tilde{\nu}_{\text{TS}} \approx 400\text{--}900 \text{ cm}^{-1} \quad (r_{\text{TS}} = 1.5\text{--}1.8 \text{ \AA}) \quad (\text{S21})$$

Enhanced backdonation (Γ_{NN} increases or holds). As the N–N bond elongates, π^* -backdonation from the Fe d-band increases, further broadening the N_2 vibrational level. The electronic friction at the TS is therefore $\geq \Gamma_{\text{NN}}^{\text{GS}} = 0.45 \text{ eV}$. Taking the conservative lower bound $\Gamma_{\text{NN}}^{\text{TS}} = \Gamma_{\text{NN}}^{\text{GS}}$:

$$\chi_{\text{TS}} \geq \frac{0.45 \text{ eV}}{2 \times 1.24 \times 10^{-4} \text{ eV} \cdot \text{cm} \times 900 \text{ cm}^{-1}} \approx 2.0 \quad (\text{S22})$$

and with enhanced backdonation and the softer TS frequency:

$$\chi_{\text{TS}} \lesssim \frac{0.70 \text{ eV}}{2 \times 1.24 \times 10^{-4} \times 400 \text{ cm}^{-1}} \approx 7.1 \quad (\text{S23})$$

Physical reconciliation

The ground-state Route C gives $\chi_{\text{DFT}} \approx 1.28$, describing the approach dynamics in the precursor well. The TS bracket places $\chi_{\text{TS}} \in [2, 7]$ at the saddle point. This is not a contradiction: the physically meaningful claim is that approach-region χ_0 governs whether the coordinate commits to the barrier or recrosses. For unpromoted Fe(111), $\chi_{\text{DFT}} \approx 0.10$ (deeply underdamped), so N_2 oscillates in the precursor well without committing and the transition state is rarely reached. For K-promoted Fe(111), $\chi_{\text{DFT}} \approx 1.28$ (near-critical approach), so N_2 approaches the barrier monotonically with minimal recrossing.

Once at the inverted saddle, the dynamics are governed by the equation $\ddot{q} + \Gamma_{\text{eff}}\dot{q} - \omega_b^2 q = 0$, whose positive reactive eigenvalue $\lambda_+ \approx \omega_b^2/\Gamma_{\text{eff}}$ for large Γ_{eff} . A large saddle-region damping ($\chi_{\text{TS}} \gg 1$) therefore means a *slower* reactive transit (λ_+ decreases as Γ_{eff} increases), not a faster one. The saddle-region χ_{TS} bracket $[2, 7]$ should be interpreted as indicating strong post-commitment friction that slows the reactive descent rather than accelerating it. This is consistent with the PGH framework: the approach-region χ_0 determines the commitment probability; the saddle-region dynamics are a separate transit-time correction. The PGH unified rate formula identifies the approach damping ratio, not the post-TS value, as what governs the rate turnover.

Direct experimental test

If the TS frequency is measured directly via reaction-path normal-mode analysis from the Nosir PES or via ML dynamics, the calculation above becomes exact. The predicted value is $\tilde{\nu}_{\text{TS}} \approx 400\text{--}900 \text{ cm}^{-1}$, corresponding to $\chi_{\text{TS}} \approx 2\text{--}7$. A measurement in this range confirms the prediction; a measurement returning $\tilde{\nu}_{\text{TS}} > 1200 \text{ cm}^{-1}$ (requiring near-zero bond softening) would be inconsistent with the known TS geometry and would require revision.

S3.4 PCET: mechanism selection via χ_{eff}

In PCET, the coupled proton–electron coordinate has an effective dissipation including both solvent dielectric friction on the electron and coupling of the proton motion to bath phonons:

$$\chi_{\text{eff}} = \frac{\Gamma_{\text{ET}} + \Gamma_{\text{PT}}}{2\Omega_{\text{coupled}}} \quad (\text{S24})$$

$\text{H} \rightarrow \text{D}$ substitution increases m_{PT} by a factor of two. Because $\Omega \propto m^{-1/2}$, this *lowers* Ω_{PT} by $\sqrt{2}$ in the proton-dominated limit:

$$\Omega_D = \frac{\Omega_H}{\sqrt{2}} \implies \chi_{\text{eff}}^D = \sqrt{2} \chi_{\text{eff}}^H \approx 1.41 \chi_{\text{eff}}^H \quad (\text{upper bound; proton-dominated}) \quad (\text{S25})$$

This result applies when $m_{\text{PT}} \approx m_{\text{proton}}$. In PCET systems where collective solvent reorganization contributes significantly to the effective coordinate mass ($m_{\text{eff}} \gg m_{\text{proton}}$), the actual frequency shift upon

H→D substitution is smaller than $1/\sqrt{2}$, and $\chi_{\text{eff}}^D < \sqrt{2}\chi_{\text{eff}}^H$. The factor $\sqrt{2}$ is therefore an upper bound; the qualitative direction (deuteration raises χ_{eff}) is robust.

If $\chi_{\text{eff}}^H \approx 1$ (concerted), then $\chi_{\text{eff}}^D \approx 1.41$ (overdamped, spatial-diffusion limited). The deuterated system is pushed toward the overdamped regime, making the mechanism sluggish and stepwise. Conversely, if $\chi_{\text{eff}}^H \approx 0.7$, then $\chi_{\text{eff}}^D \approx 0.99$, placing the deuterated coordinate closer to the critical window than the protiated system. The experimental signature is a KIE that rises then falls with increasing friction, reaching its maximum when χ_{eff}^H approaches the critical window from below (Test 4, Section S4). This KIE maximum is a dynamical signature of the $\chi = 1$ separatrix, not a tunneling signature, and is predicted in a fully classical framework.

S4 Falsification Protocol

The damping descriptor developed here makes experimentally testable predictions within the stated validity domain. Each test in Table S5 defines experimental conditions, minimum dataset requirements, predicted outcomes, and a clear falsification criterion. A single well-controlled failure within the stated domain and scope falsifies the corresponding claim.

#	Domain	Measurement	Prediction	Falsified if	Min. dataset
1	ET	k_{et} vs. viscosity at fixed λ , ΔG°	Rate maximum at viscosity where $\chi_{\text{ET}} \approx 1$	Rate monotonic in viscosity; no maximum	≥ 5 viscosities; λ constant $\pm 5\%$
2	ET	Femtosecond TA vs. solvent across $\chi = 0.1\text{--}2$	$\chi < 1$: oscillatory wavepacket; $\chi \approx 1$: spectral merging; $\chi > 1$: monotonic	No qualitative spectral transition near $\chi = 1$	≥ 3 solvents spanning full range
3	PCET	KIE, pH, intermediates vs. pressure 0–10 kbar	Stepwise $\rightarrow$ concerted switch expected when χ_{eff} approaches the critical window $[0.8, 1.3]$	Mechanism switch uncorrelated with χ	≥ 5 pressures; χ computed at each P
4	PCET	k_H/k_D vs. solvent friction at fixed T	KIE non-monotonic (rises then falls); feature when χ_{eff}^H or χ_{eff}^D approaches near-critical window; D shifted by $\sqrt{2}$ (overdamped direction)	KIE friction-independent or strictly monotonic	≥ 4 solvents; H and D measured separately
5	Catalysis	Volcano-plot outlier TOF vs. χ_{spec} or χ_{DFT} across ≥ 5 metals/facets	Positive outliers at $\chi \approx 1$; negative outliers at $\chi \ll 1$ or $\gg 1$	Outlier activity correlates with binding energy alone	≥ 5 metals or facets with χ and TOF
6	Catalysis	$^{14}\text{N}/^{15}\text{N}$ dissociation ratio on Fe(111) vs. Fe(110)	Fe(111) shows larger isotope effect (mass shift moves χ further from window)	Isotope effects surface-independent	Both facets; both isotopes; same conditions
7	Catalysis	Operando $\Delta\tilde{\nu}_L/\tilde{\nu}$ and TOF vs. K-promoter loading	TOF maximizes at finite loading where $\chi_{\text{spec}} \approx 1$; over-promotion reduces TOF	TOF increases monotonically with $\Delta\tilde{\nu}_L/\tilde{\nu}$	≥ 3 promoter loadings; TOF $\pm 10\%$; Voigt-fitted linewidths
8	Cross-domain	χ at optimal conditions across ET, PCET, catalysis for independent systems	All optimal systems cluster within $\chi \in [0.8, 1.3]$; $p < 0.05$ vs. uniform prior on $\log_{10} \chi \in [-4, 1]$	Optimal χ scatter outside $[0.8, 1.3]$; no clustering	≥ 1 independent optimal system per domain

Table S5: Eight falsifiable predictions. All tests must be performed within scope boundaries (thermally activated, near-Markovian, KIE < 7). Test 7 is the most experimentally direct original prediction.

Test 7: Route A validity condition

Test 7 requires that the Lorentzian fraction of the operando N–N stretch linewidth exceed 50%; if Gaussian broadening dominates at operating temperature, Route A is inapplicable and the linewidth measurement must first be corrected or the test deferred to lower-temperature model surface conditions before a χ_{spec} estimate can be interpreted against the TOF data.

Test 8: independence requirement and statistical note

Test 8 computes $p = (w/r)^N$ where $w = \log_{10}(1.3/0.8)$ and $r = \log_{10}(10^1/10^{-4}) = 5$ (log-uniform prior on accessible χ -space). This interval spans the physically accessible range from extremely underdamped molecular dynamics ($\chi \approx 10^{-4}$, e.g. gas-phase N_2) to strongly overdamped condensed-phase kinetics ($\chi \approx 10^1$, e.g. heavy-metal surface adsorbates), and assumes no prior knowledge of where optima should fall. For $N = 3$ independent systems: $p \approx (0.0422)^3 \approx 7.5 \times 10^{-5}$.

Robustness to prior choice. A sweep over prior bounds from $[10^{-5}, 10^1]$ to $[10^{-4}, 10^3]$ and window widths from $[0.7, 1.3]$ to $[0.85, 1.25]$ yields $p \in [1.4 \times 10^{-5}, 2.2 \times 10^{-4}]$, with $p < 0.05$ in every combination tested. The qualitative clustering conclusion is therefore not sensitive to the specific prior choice.

Independence of the three systems. The three qualifying systems span chemically and mechanistically distinct families: (1) solvent-controlled outer-sphere electron transfer (ET), where damping enters through dielectric relaxation; (2) coupled proton–electron transfer (PCET), where damping enters through vibronic bath coupling; and (3) metal-surface catalysis, where damping enters through electronic hybridization. These systems share no common substrate, measurement technique, or adjustable parameter. Multiple solvents from the same donor–acceptor pair, or multiple promoter loadings on the same surface, do not qualify as independent systems for this test.

Current evidentiary status. Only the catalysis case provides a direct numerical χ estimate ($\chi_{\text{DFT}} \approx 1.28$). The ET anchor rests on the PGH structural argument (within the thermally activated near-Markovian regime, systems with a rate turnover are expected to place the maximum near $\chi \approx 1$ by construction), and the PCET anchor rests on the near-zero activation volume as a mechanistic proxy rather than a computed χ value. Test 8 is therefore *prospective*: the statistical calculation demonstrates the sensitivity of the framework and the significance that would be achieved if all three domains were directly numerically calibrated, but it does not yet constitute a formal statistical validation.

The three current candidate anchors are: (1) the ET domain, where the PGH structural argument (Section S1.2) establishes that any system exhibiting a Kramers rate turnover has its maximum near $\chi \approx 1$ by construction; the TEMPO series on the overdamped flank is consistent with this; (2) the PCET domain, identified through the near-zero activation volume $\Delta V^\ddagger$ in the concerted regime (Ref. [28] of main text), which confirms friction bypass at the near-critical damping condition; (3) the catalysis domain, $\chi_{\text{DFT}} \approx 1.28$ for K-promoted Fe(111) computed from first-principles electronic friction tensors via Route C. Test 8 becomes a formal statistical validation when at least one direct numerical χ estimate is available in each domain. As such estimates accumulate, N grows and the prospective p -value strengthens accordingly.

S5 Computational Implementation

The following Python implementation uses the standard library only (no external dependencies). All physical constants are NIST 2018 CODATA values. This script is the canonical computational artifact deposited at Zenodo.

Listing 1: `symc_methods.py` — computational core for the damping-ratio estimators

```
"""
symc_methods.py
Damping-ratio computational core (standard library only).
Author: Nate Christensen, SymC Universe Project, March 2026
"""
import math

# NIST 2018 CODATA constants
EV_TO_J    = 1.602176634e-19    # exact (2019 SI redefinition)
AMU_TO_KG  = 1.66053906660e-27
HC_EV_CM   = 1.23984193e-4     # hc in eV*cm

def chi_route_a(delta_nu_L_cm1: float, nu_cm1: float) -> float:
    """Route A: spectroscopic. delta_nu_L MUST be Lorentzian-only component
    ."""
    if delta_nu_L_cm1 <= 0 or nu_cm1 <= 0:
        raise ValueError("Frequencies must be positive.")
    return delta_nu_L_cm1 / (2.0 * nu_cm1)

def chi_route_b_relative(tau_l_a_ps: float, tau_l_b_ps: float,
                        omega_s_a_cm: float = None,
                        omega_s_b_cm: float = None) -> float:
    """Route B (relative): chi_A / chi_B approx tau_L_A / tau_L_B.
    Physical basis: Gamma = C_solv * tau_L; C_solv = Omega_s^2.
    m_eff, lambda, DQ cancel exactly for fixed donor-acceptor pair.
    C_solv cancels approximately for same-class solvents (Omega_s similar).
    For cross-class comparisons, supply omega_s_a_cm and omega_s_b_cm to
    apply the (Omega_s_A/Omega_s_B)^2 correction explicitly."""
    ratio = tau_l_a_ps / tau_l_b_ps
    if omega_s_a_cm is not None and omega_s_b_cm is not None:
        ratio *= (omega_s_a_cm / omega_s_b_cm) ** 2
    return ratio

def chi_route_b_absolute(tau_l_ps: float, lambda_ev: float,
                        delta_q_ang: float, m_eff_amu: float,
                        omega_s_cm: float) -> float:
    """Route B (absolute): chi = (omega_s^2 * tau_L) / (2 * Omega_0).
    Physical basis: Gamma = omega_s^2 * tau_L (van der Zwan & Hynes 1982).
    omega_s: solvent librational frequency (typically 50-150 cm^-1).
    ALWAYS call over a range of m_eff_amu and omega_s. Report full bracket
    ."""
    tau_s    = tau_l_ps * 1e-12
    lam_j    = lambda_ev * EV_TO_J
```

```

dq_m      = delta_q_ang * 1e-10
m_kg      = m_eff_amu * AMU_TO_KG
omega_0    = math.sqrt(2.0 * lam_j / (m_kg * dq_m**2))      # rad/s
omega_s    = omega_s_cm * 2 * math.pi * 2.99792458e10      # rad/s (NIST
    )
gamma      = omega_s**2 * tau_s
return gamma / (2.0 * omega_0)

def chi_route_c(gamma_cat_ev: float, nu_cm1: float,
                gamma_int_ev: float = 0.0) -> float:
    """Route C: DFT electronic friction.
    gamma_cat from mode-projected electronic friction tensor."""
    return (gamma_cat_ev + gamma_int_ev) / (2.0 * HC_EV_CM * nu_cm1)

def chi_pcet_isotope(chi_H: float) -> float:
    """PCET isotope shift H -> D.
    H->D doubles mass, lowers Omega by 1/sqrt(2), raises chi by sqrt(2).
    chi_D = sqrt(2) * chi_H (overdamped direction)."""
    return math.sqrt(2.0) * chi_H

def test8_p_value(n_independent: int,
                  w_min: float = 0.8, w_max: float = 1.3,
                  log_min: float = -4.0, log_max: float = 1.0) -> dict:
    """Test 8 p-value. n_independent MUST be genuinely independent systems
    ."""
    log_range = log_max - log_min
    window_frac = math.log10(w_max) - math.log10(w_min)
    p_single = window_frac / log_range
    log10_p = n_independent * math.log10(p_single)
    return {"p_single_pct": round(p_single * 100, 4),
            "log10_p": round(log10_p, 3),
            "p_exact": 10 ** log10_p,
            "significant": log10_p < math.log10(0.05)}

# Verification and sensitivity analysis
if __name__ == "__main__":
    print("=== Route B: relative chi, TEMPO solvent series ===")
    solvents = [("Acetonitrile", 0.20), ("Water", 0.20), ("D2O", 0.30),
                ("THF", 0.80), ("Propylene carbonate", 1.20)]
    tau_ref = 0.20
    for name, tau in solvents:
        chi_rel = chi_route_b_relative(tau, tau_ref)
        print(f" {name:<25} tau_L={tau:.2f}ps chi_rel={chi_rel:.2f}x "
              f"({'more overdamped' if chi_rel > 1 else 'reference'})")
    print(" Trend consistent with high-friction flank; absolute regime
          requires anchor.")

    print("\n=== PCET isotope shift (H -> D) ===")
    for chi_H in [0.7, 0.85, 1.0, 1.1]:
        chi_D = chi_pcet_isotope(chi_H)
        print(f" chi_H={chi_H:.2f} -> chi_D={chi_D:.3f} (raised by sqrt(2)={

```

```

        math.sqrt(2):.3f}))")

print("\n=== Route C: Fe surface series ===")
print(f"   Fe(111) K-promoted (Gamma=0.45 eV, nu=1415 cm-1): "
      f"chi = {chi_route_c(0.45, 1415):.3f}")
print(f"   Fe(110) (Gamma=0.30 eV, nu=2000 cm-1): "
      f"chi = {chi_route_c(0.30, 2000):.3f}")
print(f"   Fe(111) unpromoted (Gamma=0.05 eV, nu=2100 cm-1): "
      f"chi = {chi_route_c(0.05, 2100):.3f}")
print("   Unit check: chi = Gamma[eV] / (2 * hc[eV*cm] * nu[cm-1]) = eV/
      eV (dimensionless)")

print("\n=== Test 8: prior sensitivity sweep ===")
import math as _m
for xmin, xmax in [(1e-5,1e1),(1e-4,1e1),(1e-4,1e2),(1e-4,1e3)]:
    for lo, hi in [(0.8,1.3),(0.75,1.35),(0.7,1.4),(0.85,1.25)]:
        p1 = (_m.log(hi)-_m.log(lo)) / (_m.log(xmax)-_m.log(xmin))
        pN = p1**3
        print(f"   [{xmin:.0e},{xmax:.0e}] window[{lo},{hi}] "
              f"p_single={p1:.4f}  p_N3={pN:.2e}  sig={'yes' if pN<0.05
              else 'no'}")

print("\n=== Test 8: N=3 base case ===")
s = test8_p_value(3)
print(f"   p_single = {s['p_single_pct']:.3f}%")
print(f"   log10(p) = {s['log10_p']:.2f}")
print(f"   p_exact   = {s['p_exact']:.2e}")
print(f"   Significant (p<0.05): {s['significant']}")

```

S5.1 Reproducible relative-chi calculation for the ET solvent series

Under the corrected Route B model ($\chi \propto \tau_L$), the operationally robust quantity for the TEMPO solvent series is the relative damping ratio, computed from the relative- χ relation $\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$ using acetonitrile ($\tau_L = 0.20$ ps) as the reference.

```

import math

solvents = [
    ("Acetonitrile", 0.20),
    ("Water", 0.20),
    ("D2O", 0.30),
    ("THF", 0.80),
    ("Propylene carbonate", 1.20),
]

tau_ref = 0.20 # acetonitrile reference

for name, tau in solvents:
    chi_rel = tau / tau_ref

```

```
print(f"{name}: chi_rel = {chi_rel:.2f}x")
```

which returns

```
Acetonitrile:      chi_rel = 1.00x
Water:             chi_rel = 1.00x
D2O:               chi_rel = 1.50x
THF:               chi_rel = 4.00x
Propylene carbonate: chi_rel = 6.00x
```

The rate k_{et} decreases monotonically across the series as τ_L increases. This is consistent with sampling the high-friction (spatial-diffusion) side of the Kramers turnover under $\chi \propto \tau_L$, but absolute flank assignment requires either a full turnover scan or an independently calibrated χ anchor. The structural argument of Section S1.2 establishes that the near-critical optimum sits at $\chi \approx 1$ by the mathematical form of the PGH rate, which anchors the ET domain for Test 8 to a structural argument rather than to a specific numerical calculation.

S6 Extensions and Open Problems

S6.1 Multidimensional systems

When multiple coordinates couple to the reactive mode, the scalar χ is replaced by a mode-projected effective quantity:

$$\chi_{\text{eff}} = \frac{\sum_k |c_k|^2 \Gamma_k}{2\sqrt{\sum_k |c_k|^2 \Omega_k^2}} \quad (\text{S26})$$

This is a dominant-mode approximation that assumes one mode carries the majority of the reactive weight. Mode weights $|c_k|^2$ are used rather than c_k to ensure nonnegativity: signed or complex overlap coefficients could otherwise produce cancellation in the numerator or denominator, yielding unphysical negative or zero χ_{eff} values. The strictly rigorous expression requires a quadratic form constructed from the full friction tensor projected onto the committor subspace; Eq. (S26) is the leading-order reduction of that form when one mode dominates. The appropriate level of approximation depends on how concentrated the reactive weight is: when the dominant mode coefficient $c_{\text{max}} > 0.7$, one mode carries the reactive weight and the scalar χ_{eff} is adequate; when $0.5 < c_{\text{max}} \leq 0.7$, moderate multi-mode mixing is present and the scalar may be used with explicit qualification; when $c_{\text{max}} \leq 0.5$, strong multi-mode coupling is present and the full friction tensor projection is required. Identification of the dominant mode uses saddle-point normal-mode analysis: the mode with the largest component in the transition vector (imaginary-frequency eigenvector at the transition state).

S6.2 Machine learning integration

The damping ratio χ can be incorporated as a descriptor in catalyst screening models. Current approaches use adsorption energy (ΔE_{ads}) and d-band center (ε_d) as primary descriptors. χ is orthogonal to both: it captures dynamical commitment rather than thermodynamic binding. A three-descriptor space (ΔE_{ads} , ε_d , χ) is expected to resolve BEP outliers that neither two-descriptor combination can separate. Implementation: add χ_{DFT} from mode-projected friction tensors to the training feature set and evaluate feature

importance for outlier separation.

S6.3 Enzyme active sites

Enzyme active sites tune both Γ_{eff} through coupling to protein breathing modes and solvent channels, and Ω_{TS} through geometric strain on the substrate. The prediction is that evolved enzymes exhibit $\chi_{\text{eff}} \approx 1$ for natural substrates, with off-target substrates exhibiting χ displaced from unity. Suitable test systems: cytochrome P450 (ET step, Route B measurable) and RNase A (PCET step, Route A measurable via ultrafast vibrational spectroscopy on the P–O stretch).

S7 Constrained χ -Space and the Substrate-Damping Proxy Map

S7.1 Accessible χ -space by domain class

Not all chemical systems can access both sides of the $\chi = 1$ boundary. The accessible χ -space is substrate-determined, and this structural fact organizes why efficiency maxima appear in different locations for different reaction classes.

Domain	Accessible χ -space	Location of optimum
ET / PCET, solvent dynamics	Both phases	Near separatrix ($\chi \approx 0.8\text{--}1.2$)
Metallic surface catalysis	Substrate-dependent; spans underdamped to near-critical or overdamped depending on surface and promoter	Near best accessible commitment window for the specific surface family
Wide-gap insulators, noble gases	Underdamped only	Dissipation is not the kinetic bottleneck

Table S6: Accessible χ -space and optimum location by domain.

In electron-transfer and PCET systems, both dynamical phases are accessible by varying solvent, composition, or pressure, and efficiency maxima emerge near the phase boundary for structural reasons. Metallic surfaces occupy a more restricted but not universally overdamped region: the Fe series demonstrates ($\chi_0 \approx 0.10$ for unpromoted Fe(111), ≈ 0.61 for Fe(110), ≈ 1.28 for K-promoted Fe(111)) that the accessible range spans from underdamped through near-critical. Catalytic optimization operates within the accessible band for the specific surface family, not in a universally overdamped regime. The $\chi_0 = 1$ separatrix is a generator-level boundary common to all domains; the accessible χ_0 -range is substrate-determined.

S7.2 Standard-state lattice χ map

The lattice damping proxy $\chi_{\text{lattice}} = \pi\lambda_{ep}T/(2\Theta_D)$ is derived from the Debye-model electron-phonon coupling rate at the bath frequency $\Omega_{\text{bath}} = k_B\Theta_D/\hbar$ [21,22]. This quantity is not χ_{TS} ; it is a substrate-damping proxy that describes how strongly the phonon bath couples to an electronic reactive coordinate at the Fermi level, and therefore which region of χ -space a material inherits as its operating environment.

χ_{lattice} uses λ_{ep} (electron-phonon coupling constant) rather than phonon self-damping. The ratio χ measures how an electronic degree of freedom is damped by the phonon bath, not how phonons damp themselves. Using phonon self-damping would place Fe at $\chi \approx 0.01\text{--}0.08$ and remove all near-critical transition metals from the near-unity cluster. Using λ_{ep} , effective catalytic transition metals (Fe, Ru, Co, Ni) cluster near $\chi \approx 1$; wide-gap insulators and noble gases occupy the underdamped floor; heavy-atom metals (Hg, Pb, Tl) are strongly overdamped. The clustering emerges entirely from independently tabulated λ_{ep} and Θ_D values (McMillan 1968 [21]; Allen-Dynes 1975 [22]; standard solid-state references) and is not imposed by parameter selection. The full periodic-table visualization is Figure 3 of the main manuscript.

More broadly, these results suggest that dynamical damping may serve as a useful descriptor alongside thermodynamic quantities when interpreting reaction efficiency across chemically distinct systems.

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Note: References in this Supplementary Information are numbered independently of the main manuscript. All references cited here are also listed in the main manuscript reference list.

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Code (`symc_methods.py`) and all input parameters deposited at Zenodo (DOI: 10.5281/zenodo.17891562). Contact: NateChristensen@SymCUniverse.com.